TASK RECORD

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TASK: Linux chmod Command with Numerical Values and Letter Commands.

Task Description:

To understand and practice the Linux chmod command using both numerical values and symbolic (letter) notation to modify file permissions.

Tools / Commands:

1. chmod 777 sample.txt + ls -lart,
2. chmod 755 sample.txt + ls -lart,
3. chmod 644 sample.txt + ls -lart,
4. chmod 700 sample.txt + ls -lart.

Objectives:

The chmod (Change Mode) command in Linux is used to change the permissions of files and directories. It controls who can **read**, **write**, or **execute** a file. Permissions can be set using either **numerical values** or **symbolic letters**.

What is chmod?

Syntax:

chmod [permissions] filename

Example: 1

chmod 755 sample.txt

Example: 2

chmod u+rwx sample.txt

Where:

- **u** = User (Owner)
- **g** = Group
- **o** = Others
- **a** = All (User + Group + Others)

Permissions:

- **r** = Read
- **w** = Write
- **x** = Execute

TASK1: NUMERICAL PERMISSION VALUES.

Permission Values**:**

VALUE	PERMISSION	MEANING
0	---	No Permission
1	--x	Execute Only
2	-w-	Write Only
3	-wx	Write + Execute
4	r--	Read Only
5	r-x	Read + Execute
6	rw-	Read + Write
7	rwx	Read + Write + Execute

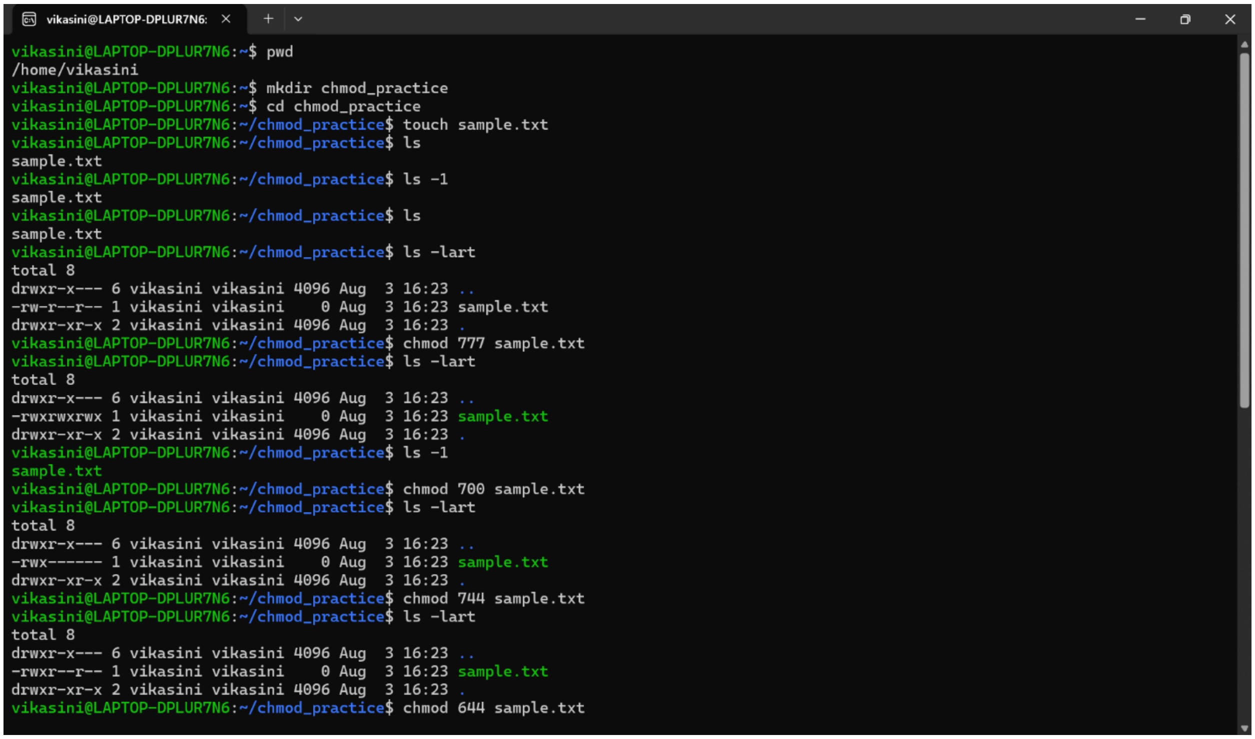
OUTPUT:

```
vikasini@LAPTOP-DPLUR7N6:~$ cd chmod_practice
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ touch sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
-rwxrwxrwx 1 vikasini vikasini    0 Aug  3 17:31 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ^C
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$  chmod 777 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
-rwxrwxrwx 1 vikasini vikasini    0 Aug  3 17:31 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 700 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
-rwx------ 1 vikasini vikasini    0 Aug  3 17:31 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 744 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
-rwxr--r-- 1 vikasini vikasini    0 Aug  3 17:31 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$
```

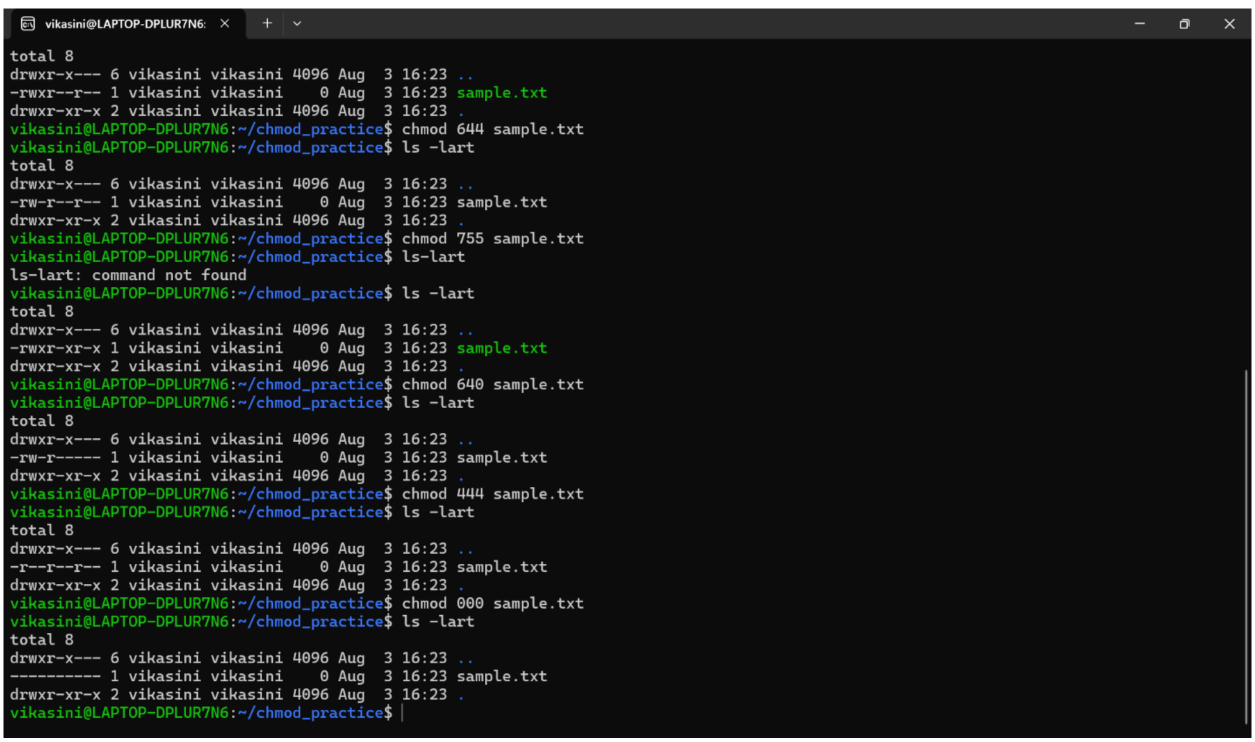
COMMON NUMERICAL PERMISSIONS:

Number	Letter Format	Meaning
777	rwxrwxrwx	Full permission for everyone
755	rwxr-xr-x	Owner: Full, Group: Read & Execute, Others: Read & Execute
700	rwx------	Owner: Full only
644	rw-r--r--	Owner: Read & Write, Others: Read only
600	rw-------	Owner: Read & Write only
444	r--r--r--	Owner: Read & Write only
000	---------	Owner: Read & Write only

OUTPUT:



```
vikasini@LAPTOP-DPLUR7N6:~$ pwd
/home/vikasini
vikasini@LAPTOP-DPLUR7N6:~$ mkdir chmod_practice
vikasini@LAPTOP-DPLUR7N6:~$ cd chmod_practice
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ touch sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls
sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -1
sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls
sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rw-r--r-- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 777 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rwxrwxrwx 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -1
sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 700 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rwx------ 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 744 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rwxr--r-- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 644 sample.txt
```



```
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rwxr--r-- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 644 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rw-r--r-- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 755 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls-lart
ls-lart: command not found
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rwxr-xr-x 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 640 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rw-r----- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 444 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-r--r--r-- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 000 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
---------- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$
```

TASK2: SYMBOLIC (LETTER) COMMANDS:

Command	Command
chmod u+x	Add execute permission to owner
chmod u-w	Remove write permission from owner
chmod g+r	Add read permission to group
chmod o-r	Remove read permission from others
chmod a+rw	Give read & write permission to everyone
chmod a-x	Remove execute permission from everyone
chmod u=rwx	Set owner permission to read, write and execute
chmod g=r	Set group permission to read only
chmod o=	Remove all permissions for others

TRICK:

1. **7 = rwx** (4+2+1)
2. **6 = rw-** (4+2)
3. **5 = r-x** (4+1)
4. **4 = r--** (4)
5. **3 = -wx** (2+1)
6. **2 = -w-** (2) ,**1 = --x** (1)
7. **0 = ---** (No permission).

OUTPUT:

```
vikasini@LAPTOP-DPLUR7N6:
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod u+x sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
---x------ 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod u-w sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
---x------ 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod g+r sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
---xr----- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod o-r sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
---xr----- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod a-x sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
----r----- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod a+rw sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rw-rw-rw- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
```

```
vikasini@LAPTOP-DPLUR7N6:
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rwxr--r-- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 644 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rw-r--r-- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 755 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls-lart
ls-lart: command not found
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rwxr-xr-x 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 640 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-rw-r----- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 444 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
-r--r--r-- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ chmod 000 sample.txt
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$ ls -lart
total 8
drwxr-x--- 6 vikasini vikasini 4096 Aug  3 16:23 ..
---------- 1 vikasini vikasini    0 Aug  3 16:23 sample.txt
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 .
vikasini@LAPTOP-DPLUR7N6:~/chmod_practice$
```

CONCLUSION:

The Linux chmod command is used to manage file and directory permissions. Numerical values (0–7) provide a quick way to assign permissions, while symbolic notation (u, g, o, a, r, w, x) allows individual permission changes. Understanding both methods helps in securely managing Linux files and directories.